

JENNIFER WINTER, PHD

Department of Economics and School of Public Policy
University of Calgary Downtown Campus
906 8th Avenue S.W., 5th floor, Calgary, Alberta T2P 1H9

Phone: 403-210-7889
Email: jwinter@ucalgary.ca

Web: www.jenniferwinter.ca | <https://profiles.ucalgary.ca/jennifer-winter>

Socials: X: [@dr_jen_winter](https://twitter.com/dr_jen_winter) Bluesky: [@drjenwinter.bsky.social](https://bsky.app/profile/drjenwinter.bsky.social) Threads: [@dr.jennifer.winter](https://www.threads.net/@dr.jennifer.winter)

This version: November 2024; latest version available [here](#).

PROFESSIONAL EXPERIENCE

2024 – Professor, Department of Economics and School of Public Policy, University of Calgary
2023 – Departmental Science Advisor, Environment and Climate Change Canada, Government of Canada (*60% interchange*)
2020 – 2024 Associate Professor, Department of Economics and School of Public Policy, University of Calgary
2022 – 2023 Visiting Scholar, Environment and Climate Change Canada (*sabbatical*)
2016 – 2022 Scientific Director, Energy and Environmental Policy, School of Public Policy, University of Calgary
2017 – 2020 Assistant Professor (Tenure Track), Department of Economics, University of Calgary
2015 – 2017 Assistant Professor (Limited Term), Department of Economics, University of Calgary
2015 – 2018 Director, Canadian Network for Energy Policy Research and Analysis, School of Public Policy, University of Calgary
2014 – 2015 Associate Director, Energy and Environmental Policy, School of Public Policy, University of Calgary
2012 – 2014 Research Associate, Energy and Environmental Policy, School of Public Policy, University of Calgary
2011 – 2012 Research Advisor, Labour Program, Human Resources and Skills Development Canada

OTHER CURRENT AFFILIATIONS

2019 – Mitigation Expert Panel Member, Canadian Climate Institute
2017 – Member, Environmental Assessment Advisory Group, Global Affairs Canada
2017 – Treasurer, WPC Energy Canada Future Leaders
2016 – Treasurer, Canadian Resource and Environmental Economics Association
2014 – Board Member, WPC Energy Canada Future Leaders

RESEARCH INTERESTS

Energy and environmental economics, climate policy, emissions policy, innovation economics, Canadian economic history

EDUCATION

University of Calgary, PhD in Economics, 2011
DISSERTATION: “Essays on Government Policy and the Development of Natural Resources”
University of Calgary, MA in Economics, 2007
University of Calgary, BA in Economics (First Class Honours), 2005

RESEARCH

Refereed Journal Articles

- Winter, Jennifer, Brett Dolter & G. Kent Fellows. 2023. "Carbon Pricing Costs for Households and Revenue Recycling Options in Canada." *Canadian Public Policy* 49(1): 13-45; <https://doi.org/10.3138/cpp.2022-036>. Winner of the [2023 Vanderkamp Prize](#) for the best article in *Canadian Public Policy*.
- Fellows, G. Kent, Jennifer Winter & Alaz Munzur. 2023. "An Analysis of Industrial Policy Mechanisms to Support Commercial Deployment of Bitumen Partial Upgrading in Alberta." *Energies* 16(6): 2670; <https://doi.org/10.3390/en16062670>.
- Schaufele, Brandon & Jennifer Winter. 2023. "Production Controls in North American Heavy Oil and Bitumen Markets: Surplus Transfer Due to Alberta's Curtailment Policy." *Energies* 16(3): 1389; <https://doi.org/10.3390/en16031389>.
- Dobson, Sarah, Victoria Goodday & Jennifer Winter. 2023. "If It Matters, Measure It: A Review of Methane Sources and Mitigation Policy in Canada." *International Review of Environmental and Resource Economics* 16(3-4): 309-429. <http://dx.doi.org/10.1561/101.00000146>.
- Fellows, Kent, Alaz Munzur & Jennifer Winter. 2022. "A Socio-economic Review of the Impacts of Northwest Territories' Inuvik to Tuktoyaktuk Highway 10." *Canadian Journal of Regional Science*: 45(3): 137-149. https://idjs.ca/images/rcsr/archives/V45N3_2-FELLOWS-MUNZUR-WINTER.pdf.
- Salamati, Mehdi, Xin Wang, Jennifer Winter & Hamidreza Zareipour. 2022. "Optimal Routing of Wide Multi-Modal Energy and Infrastructure Corridors." *International Journal of Geo-Information* 11(8): 434. <https://doi.org/10.3390/ijgi11080434>.
- Tombe, Trevor & Jennifer Winter. 2021. "Fiscal Integration with Internal Trade: Quantifying the Effects of Equalizing Transfers." *Canadian Journal of Economics* 54(2): 1-35. <https://onlinelibrary.wiley.com/doi/full/10.1111/caje.12491>.
- Rioux, Jean-Sebastien & Jennifer Winter. 2020. "Forks in the Road: Energy Policies in Canada and the US since the Shale Revolution." *The American Review of Canadian Studies* 50(1): 66-85. Invited contribution. <https://doi.org/10.1080/02722011.2020.1743015>.
- Tombe, Trevor & Jennifer Winter. 2015. "Environmental Policy and Misallocation: The Productivity Effect of Intensity Standards," *Journal of Environmental Economics and Management* 72: 137-163, July. <https://doi.org/10.1016/j.jeem.2015.06.002>.

Non-refereed Journal Articles

- Winter, Jennifer. 2024. "Exploring the Landscape of Canadian Climate Policy," *Canadian Public Policy* 50(S1): 73-102. Invited contribution. <https://utpjournals.press/doi/full/10.3138/cpp.2023-055>.
- Winter, Jennifer. 2023. "The Challenge of Border Carbon Adjustments as a Mechanism for Climate Clubs." *PLOS Climate* 2(2): e0000135. Invited contribution. <https://doi.org/10.1371/journal.pclm.0000135>.
- Winter, Jennifer. 2020. "Carbon Pricing in a Federal State: The Case of Canada." *ifo DICE Report - Journal for Institutional Comparisons* 18(1): 13-19. Invited contribution. <https://www.ifo.de/en/publikationen/2020/article-journal/carbon-pricing-federal-state-case-canada>.

Peer-Reviewed Policy Articles

- Winter, Jennifer & Trevor Tombe. 2024 (in press). "Climate Policy and Affordability in Canada: Interactions and Policy Levers." Institute for Research on Public Policy.
- Galley, Emily, Katharina Koch, G. Kent Fellows, Robert Mansell, Nicole Pinto & Jennifer Winter. 2023. "The Canadian Northern Corridor Community Engagement Program: Results and Lessons Learned," *The School of Public Policy Publications* 16 (September). <https://doi.org/10.11575/sppp.v16i1.77585>.
- Winter, Jennifer, Megan Bailey, Emily Galley, Chris Joseph & Blake Shaffer. 2021. "A Multiple Account Benefit-Cost Analysis of Coal Mining in Alberta." *The School of Public Policy Publications* 14 (November). <https://doi.org/10.11575/sppp.v14i1.73574>.
- Fellows, G. Kent, Victoria Goodday & Jennifer Winter. 2021. "A Review of Barriers to Full-Scale Deployment of Emissions-Reduction Technologies." *The School of Public Policy Publications* 14 (April). <http://dx.doi.org/10.11575/sppp.v14i.70772>.
- Winter, Jennifer, Victoria Goodday & G. Kent Fellows. 2019. "Enabling Partial Upgrading in Alberta: A Review of the Regulatory Framework and Opportunities for Improvement," *The School of Public Policy Publications* 12 (December). <https://doi.org/10.11575/sppp.v12i0.68199>.
- Dobson, Sarah, Jennifer Winter & Brendan Boyd. 2019. "The Greenhouse Gas Emissions Coverage of Carbon Pricing Instruments for Canadian Provinces." *The School of Public Policy Publications* 12 (February). <https://doi.org/10.11575/sppp.v12i0.53155>.
- Dobson, Sarah & Jennifer Winter. 2018. "Assessing Policy Support for Emissions Intensive and Trade Exposed Industries." *The School of Public Policy Publications* 11 (October). <https://doi.org/10.11575/sppp.v11i0.43673>.
- Winter, Jennifer, Sarah Dobson, G. Kent Fellows, Dexter Lam & Paul Craig. 2018. "The Potential for Canadian LNG Exports to Europe." *The School of Public Policy Publications* 11 (July). <https://doi.org/10.11575/sppp.v11i0.53162>.
- Dobson, Sarah, G. Kent Fellows, Trevor Tombe & Jennifer Winter. 2017. "The Ground Rules for Effective OBAs: Principles for Addressing Carbon-Pricing Competitiveness Concerns through the Use of Output-Based Allocations." *The School of Public Policy Publications* 10 (June). <http://dx.doi.org/10.11575/sppp.v10i0.42633>.
- Fellows, G. Kent, Robert Mansell, Ronald Schlenker & Jennifer Winter. 2017. "Public-Interest Benefit Evaluation of Partial-Upgrading Technology." *The School of Public Policy Publications* 10 (January). <http://dx.doi.org/10.11575/sppp.v10i0.42615>.
- Colton, John, Kenneth Corscadden, Stewart Fast, Monica Gattinger, Joel Gehman, Martha Hall Findlay, Dylan Morgan, Judith Sayers, Jennifer Winter, & Adonis Yatchew. 2016. "Energy Projects, Social Licence, Public Acceptance and Regulatory Systems in Canada: A White Paper." *The School of Public Policy Publications* 9 (May). <http://dx.doi.org/10.11575/sppp.v9i0.42589>.
- Dobson, Sarah & Jennifer Winter. 2015. "The Case for a Carbon Tax in Alberta." *The School of Public Policy Publications* 8 (November). <http://dx.doi.org/10.11575/sppp.v8i0.42549>.
- Moore, Michal C., David Hackett, Leigh Noda, Jennifer Winter, Roman Karski, & Mark Pilcher. 2014. "Risky Business: The Issue of Timing, Entry and Performance in the Asia-Pacific LNG Market." *The School of Public Policy Publications* 7 (July). <http://dx.doi.org/10.11575/sppp.v7i0.42470>.

- Moore, Michal C., André Turcotte, & Jennifer Winter. 2014. "Aboriginal-Canadians and Energy Literacy: A Survey of Opinions and Thoughts on Energy." *The School of Public Policy Publications* 7 (February). <http://dx.doi.org/10.11575/sppp.v7i0.42458>.
- Winter, Jennifer & Michal C. Moore. 2013. "The 'Green Jobs' Fantasy: Why the Economic and Environmental Reality Can Never Live Up to the Political Promise." *The School of Public Policy Publications* 6 (October). <http://dx.doi.org/10.11575/sppp.v6i0.42444>.
- Tombe, Trevor & Jennifer Winter. 2013. "The Importance of Policy Neutrality for Lowering Greenhouse Gas Emissions." *The School of Public Policy Publications* 6 (March). <http://dx.doi.org/10.11575/sppp.v6i0.42427>.
- Moore, Michal C., André Turcotte, Jennifer Winter & P. Bernard Walp. "Energy and Energy Literacy in Canada: A Survey of Business and Policy Leadership." *The School of Public Policy Publications* 6 (February). <http://dx.doi.org/10.11575/sppp.v6i0.42421>.
- Hackett, David, Leigh Noda, Susan W. Grissom, Michal C. Moore & Jennifer Winter. 2013. "Pacific Basin Heavy Oil Refining Capacity." *The School of Public Policy Publications* 6 (February). <http://dx.doi.org/10.11575/sppp.v6i0.42417>.
- Turcotte, André, Michal C. Moore & Jennifer Winter. 2012. "Energy Literacy in Canada." *The School of Public Policy Publications* 5 (October). <http://dx.doi.org/10.11575/sppp.v5i0.42402>.
- Mansell, Robert, Jennifer Winter, Matt Krzepkowski & Michal C. Moore. 2012. "Size, Role and Performance in the Oil and Gas Sector." *The School of Public Policy Publications* 6 (February). <http://dx.doi.org/10.11575/sppp.v5i0.42395>.

Book Chapters and Monographs

- Koch, Katharina, Alaz Munzur & Jennifer Winter. 2025 (in press). "Climate Policy as a Geopolitical Tool: How the European Union's Carbon Border Adjustment Mechanism Affects its Relationships with Africa and China." Invited chapter in *The Palgrave Handbook on China-Europe-Africa relations*. Yichao Li, Francisco José Leandro, Jorge Tavares da Silva and Carlos Rodrigues, eds. Palgrave Macmillan. ISBN: 978-981-97-5639-1. <https://link.springer.com/book/9789819756391>.
- Winter, Jennifer & Brendan Boyd. 2024. *Protest and Partnership: Case Studies of Indigenous Peoples, Consultation and Engagement, and Resource Development in Canada* (co-edited book). University of Calgary Press. <https://press.ucalgary.ca/books/9781773852041/>.
- Boyd, Brendan, Sophie Lorefice & Jennifer Winter. 2024. "Indigenous, Industry and Government Perspectives on Consultation in Resource Development." Chapter 1 (pp. 23-58) in *Protest and Partnership: Case Studies of Indigenous Peoples, Consultation and Engagement, and Resource Development in Canada*, Jennifer Winter and Brendan Boyd, eds. University of Calgary Press. <https://press.ucalgary.ca/books/9781773852041/>.
- Munzur, Alaz, Katharina Koch & Jennifer Winter. 2023. "Geopolitical Implications of the EU's Carbon Border Adjustment Mechanism." Chapter 9 (pp. 199-226) in *Changing the Paradigm of Energy Geopolitics: Security, Resources and Pathways in Light of Global Challenges*, João Simões, Francisco Leandro, Eduardo Caetano de Sousa and Roopinder Oberoi, eds. Peter Lang Publishers. ISBN: 9781433191350. <https://www.peterlang.com/document/1298592>.

- Winter, Jennifer. 2022. “Mercados de emisiones y comercio internacional [Emissions Markets and International Trade].” Invited chapter in *Energy and Geostrategy*, 9th Edition. Ministerio De Defensa Press, Spain. https://www.enerclub.es/libros/Book_EG22/.
- Bataille, Chris, Sarah Dobson, Anna Kanduth & Jennifer Winter. 2021. “Alberta in Canada’s Net Zero Future: Seizing Opportunities While Adapting to Change.” Invited chapter in *Alberta’s Economic and Fiscal Future*, Robert Mansell and Ken McKenzie, eds. The School of Public Policy, University of Calgary. <https://www.policyschool.ca/wp-content/uploads/2021/11/Albertas-Economic-and-Fiscal-Future.pdf>.
- Winter, Jennifer. 2019. “Environmental Policy Transformations.” Invited chapter in *Policy Transformation in Canada: Is Past Prologue?*, Peter John Loewen, Carolyn Hughes Tuohy, Andrew Potter and Sophie Borwein, eds. University of Toronto Press. <https://utorontopress.com/9781487523244/policy-transformation-in-canada/>.
- Winter, Jennifer. 2019. “Canada’s Role in Global Energy Markets.” Invited chapter in *Energy and Geostrategy*, 6th Edition. Ministerio De Defensa Press, Spain. <https://www.enerclub.es/libros/BookEnergeo2019/>.
- Winter, Jennifer. In press. “Canada: Tensions between energy and GHG policies.” Invited chapter in *Meeting the Paris Mandate: A Cross-national Comparison of Energy Policy-making*, Patrice Geoffron, Lorna A. Greening, and Raphael Heffron, eds. Springer.

Technical Reports

- Winter, Jennifer, William Scott, Andrew Pape-Salmon, Alaz Munzur & Rahim Mohamed. 2023. “Examining the Interplay between Building Codes and Canadian FPT Policies, Programs, and Targets Related to Reducing GHG Emissions.” Report prepared for the National Research Council of Canada. May.
- Winter, Jennifer, Sarah Dobson, Gregory Galay, Julia Jachimowicz & W. Riley Sample. 2022. “Literature Review of Theories and Concepts related to Environmental Regulation.” Report prepared for the Government of British Columbia, Ministry of Environment and Climate Change Strategy. March.
- Olewiler, Nancy & Jennifer Winter. 2021. “A Review and Assessment of the Natural Gas Royalty System in British Columbia.” Report prepared for the Government of British Columbia. October.
- Fellows. G. Kent, Jennifer Winter & Alaz Munzur. 2021. “An Analysis of Industrial Policy Mechanisms to Support Commercial Deployment of Bitumen Partial Upgrading in Alberta.” Report prepared for Alberta Innovates. May.
- Goodday, Victoria, Jennifer Winter & Alana Westwood. 2020. “Public-Interest Determination for Infrastructure Development: A Review of Guidance and Practice in Canada.” Knowledge Synthesis Report prepared for the Social Sciences and Humanities Research Council. April.
- Shaffer, Blake & Jennifer Winter. 2020. “Defining and Describing Energy Poverty in British Columbia: The Distribution of Households' Energy Expenditure.” Report prepared for the BC Basic Income Panel. December.
- Fellows, G. Kent, Victoria Goodday & Jennifer Winter. 2019. “A Review of Barriers to Full-Scale Deployment of Emissions-Reduction Technologies.” Report prepared for Emissions Reduction Alberta. December. (*Subsequently published.*)

- Winter, Jennifer & Akash Asif. 2018. "Comprehensive Assessment of Regulatory Programs." Report to the Treasury Board of Canada Secretariat. June.
- Boyd, Brendan, Jessie Arthur & Jennifer Winter. 2018. "Meaningful Engagement of Indigenous Peoples and Local Communities in Marine Activities: Phase 2 Analysis." Report to the PAME Working Group of the Arctic Council. March.
- Boyd, Brendan, Sophie Lorefice & Jennifer Winter. 2017. "Indigenous, Industry and Government Perspectives on Consultation in Resource Development," Knowledge Synthesis Report prepared for the Social Sciences and Humanities Research Council. September.
- Winter, Jennifer. 2017. "The Effect of Carbon Pricing on Canadian Households." Briefing Note to the Senate of Canada, Standing Committee on Energy, the Environment and Natural Resources.
- Winter, Jennifer, Sarah Dobson & Sophie Lorefice. 2016. "Hydraulic Fracturing and Public Policy." Knowledge Synthesis Report prepared for the Social Sciences and Humanities Research Council. May.

Policy Papers and Reports (not peer-reviewed)

- Winter, Jennifer. 2023. "Comment on 'Reducing Canada's landfill methane emissions: Proposed regulatory framework'." *The School of Public Policy Publications* (Briefing Paper), 16 (November). <https://doi.org/10.11575/sppp.v16i1.77619>.
- Dolter, Brett & Jennifer Winter. 2022. "Electricity and equity in Canada's net zero transition: Options for rate design and electricity system funding." Scoping paper, Canadian Climate Institute. <https://climateinstitute.ca/wp-content/uploads/2022/09/Electricity-and-equity-canadas-energy-transition.pdf>.
- Winter, Jennifer. 2022. "Cities, Emissions and Climate Change." Who Does What Series: The Municipal Role in Climate Policy, Institute for Municipal Finance and Governance, University of Toronto. <https://munkschool.utoronto.ca/imfg/report/climate-policy/>.
- Schaufele, Brandon & Jennifer Winter. 2021. "Measuring the Economic Impact of Alberta's Crude Oil Curtailment Policy." Policy Brief, Ivey Energy Policy and Management Centre. https://www.ivey.uwo.ca/media/3793185/iveyenergycentre_policybrief_dec2020_measuringeconimp_actalbertacrudeoil_curtailment_2-9.pdf.
- Winter, Jennifer, Sarah Dobson, G. Kent Fellows, Dexter Lam & Paul Craig. 2018. "An Overview of Global Liquefied Natural Gas Markets and Implications for Canada." *The School of Public Policy Publications* (Briefing Paper) 11 (July). <https://doi.org/10.11575/sppp.v11i0.53162>.
- Winter, Jennifer & Sarah Dobson. 2016. "Who is Getting a Carbon-Tax Rebate?" *The School of Public Policy Publications* (Communique) 9 (June). <http://dx.doi.org/10.11575/sppp.v9i0.42595>.
- Beach, Shantel, Andrew Wilkins & Jennifer Winter. 2014. "The Future of Energy Regulation and Policy Development: A Summary Paper." *The School of Public Policy Publications* (Summary Paper) 7 (July). <http://dx.doi.org/10.11575/sppp.v7i0.42472>.
- Jennifer Winter. 2014. "Safety in Numbers: Evaluating Canadian Rail Safety Data," *The School of Public Policy Publications* (Communique) 7 (April). <http://dx.doi.org/10.11575/sppp.v7i0.42913>.

Jennifer Winter. 2013. "Grasping at Straws: Comments on the Alberta Pipeline Safety Review." *The School of Public Policy Publications* (Communique) 6 (October).
<http://dx.doi.org/10.11575/sppp.v6i0.42442>.

Holburn, Guy, Patrick McNaught, M. C. Moore & Jennifer Winter. 2013. "Wind Energy in Canada: A Survey of the Policy Environment," Ivey Business School Policy Brief (June).

Working Papers

Galay, Gregory & Jennifer Winter. "No end in sight: End-of-life management of oil wells in Alberta." Resubmitted, *Resource and Energy Economics*.

Scott, William A., Jennifer Winter, Alaz Munzur & Katharina Koch. "Developing a Climate Change Mitigation Policy Inventory for Canada." Resubmitted, *Canadian Public Policy*.

Tombe, Trevor & Jennifer Winter. "From Farms to Tables: Quantifying the Effect of Emissions Pricing on Canadian Food Prices." Resubmitted, *Canadian Journal of Agricultural Economics*.

Vollrath, Coleman, Zhenyu Zhang, Chris H. Hugenholtz, Thomas E. Barchyn & Jennifer Winter. "A review of methane emissions source types and characteristics, rates, and mitigation across U.S. and Canadian cities." Submitted.

Research in Progress

Dobson, Sarah & Jennifer Winter. "EITE Definitions and Policy Coverage."

Fellows, G. Kent & Jennifer Winter. "Economic Impact Assessments: Comparing CGE to IO Models." Chapter in *Measuring the Contribution of Energy Infrastructure: A Practical Guide*, Jennifer Winter and Trevor Tombe, eds.

Fosu, Felix, Blake Shaffer & Jennifer Winter. "Measuring Energy Poverty in Canada."

Munzur, Alaz, Katharina Koch & Jennifer Winter. "Making Trade Work for Climate." Invited chapter in *The Palgrave Handbook of the Geopolitics of Energy Cooperation*.

Semeniutina, Tetiana & Jennifer Winter. "Transatlantic Cooperation in Reducing Methane Emissions from Oil and Gas: Potential Solutions to Monitoring, Measurement, Reporting and Verification Challenges." Invited chapter in *The Palgrave Handbook of the Geopolitics of Energy Cooperation*.

Shaffer, Marvin, Blake Shaffer & Jennifer Winter. "A Multiple-Account Approach to Benefit-Cost Analysis." Chapter in *Measuring the Contribution of Energy Infrastructure: A Practical Guide*, Jennifer Winter and Trevor Tombe, eds.

Stewart, Fenner, Andrew Leach & Jennifer Winter. "A Federal Backstop to Provincial Carbon Pricing: Decarbonization in an Age of Uncooperative Federalism."

Winter, Jennifer & Akash Asif. "Comprehensive Assessment of Regulatory Programs."

Winter, Jennifer, Sefik Bagdadioglu, Sarah Dobson, Kent Fellows, Sophie Lorefice & Sean Keown. "Market Opportunities for Canadian Oil."

Winter, Jennifer & Tetiana Semeniutina. "Oil and Gas Methane Regulation: Gaps and Opportunities for Improvement."

Winter, Jennifer & Trevor Tombe. *Measuring the Contribution of Energy Infrastructure: A Practical Guide* (co-edited book).

Winter, Jennifer. "History of Energy Taxes in Canada"

Other Policy Writing

Tombe, Trevor & Jennifer Winter. 2023. "Carbon pricing is not to blame for Canada's affordability challenges." *Policy Options*, December 7, 2023. Invited contribution.

<https://policyoptions.irpp.org/magazines/december-2023/carbon-price-affordability/>.

Tombe, Trevor & Jennifer Winter. 2023. "Energy and Environmental Policy Trends: Emissions Pricing and Affordability: Lessons from British Columbia." *The School of Public Policy Publications* (Policy Trends) 15 (December). <https://www.policyschool.ca/wp-content/uploads/2023/12/EE-Trends-DEC.pdf>.

Tombe, Trevor & Jennifer Winter. 2022. "Energy and Environmental Policy Trends: Making Sense of Alberta's Fuel Tax Holiday and Electricity Bill Credit." *The School of Public Policy Publications* (Policy Trends) 14 (March).

<https://journalhosting.ucalgary.ca/index.php/sppp/article/view/74943/55683>.

Goodday, Victoria, Jennifer Winter & Nicholas Schumacher. 2020. "Energy & Environmental Policy Trends: The Hidden Costs of a Single-Use Plastics Ban." *The School of Public Policy Publications* (Policy Trends) 13 (November). <https://doi.org/10.11575/sppp.v13i0.71433>.

Schumacher, Nick, Victoria Goodday, Blake Shaffer & Jennifer Winter. 2020. "Energy and Environmental Policy Trends: Cheap Renewables Have Arrived." *The School of Public Policy Publications* (Policy Trends) 13 (November). <https://doi.org/10.11575/sppp.v13i0.71383>.

Masson, Richard & Jennifer Winter. 2020. "Energy and Environmental Policy Trends: Addressing the Threat of COVID-19 and the Oil Price War in the Petroleum Sector." *The School of Public Policy Publications* (Policy Trends) 13 (March).

<https://journalhosting.ucalgary.ca/index.php/sppp/article/view/70120>.

Fellows, G. Kent, Victoria Goodday, Rabia Ladha & Jennifer Winter. 2019. "Energy and Environmental Policy Trends: Our Planet in 2040 – Comparing World Energy Outlooks." *The School of Public Policy Publications* (Policy Trends) 12 (July). <https://doi.org/10.11575/sppp.v12i0.68679>.

Tombe, Trevor & Jennifer Winter. 2019. "Energy and Environmental Policy Trends: Carbon Tax Costs Vary Widely Across Households." *The School of Public Policy Publications* (Policy Trends) 12 (March).

Winter, Jennifer & Sam Green. 2018. "Energy and Environmental Policy Trends: The Growing Opportunity for LNG in China." *The School of Public Policy Publications* (Policy Trends) 11 (June). <https://doi.org/10.11575/sppp.v11i0.53017>.

Mintz, Jack, Ronald D. Kneebone, F. L. Ted Morton & Jennifer Winter. 2014. "Policy Advice to Alberta's New Premier," *The School of Public Policy Publications* (Communique) 7 (September). doi: <https://doi.org/10.11575/sppp.v7i0.42482>.

Tombe, Trevor & Jennifer Winter. 2013. “Shouldn't all policy be science-based?” *Policy Options*, June (invited contribution). <https://policyoptions.irpp.org/magazines/nudge/shouldnt-all-policy-be-science-based/>.

Retired Working Papers

Johnson, David & Jennifer Winter. “Alberta Oil and Gas Leases: Testing Competing Institutions.”

Winter, Jennifer. “Buying Anonymity: An Investigation of Petroleum and Natural Gas Lease Auctions”

Winter, Jennifer & J. C. Herb Emery. “Estimating the Credibility of the Co-operative Commonwealth Federation’s Threat to Nationalize Oil Resources in Saskatchewan.”

RESEARCH GRANTS AND CONTRACTS

Environment and Climate Change Canada, Climate Action and Awareness Fund (\$1,896,724): 2023-2028. Co-Investigator for *Calgary Urban Methane Emissions Measurement Testbed (CURMET)*. PI: Chris Hugenholtz, University of Calgary.

Environment and Climate Change Canada, Climate Action and Awareness Fund, Academics and Think Tanks Competition (\$800,000). Principal Investigator for *Canadian Climate Policy Partnership: 2023-2026*.

Social Sciences and Humanities Research Council, Partnership Development Grant (\$200,000). Principal Investigator for *Canadian Climate Policy Partnership: 2023-2026*.

University of Calgary, Vice President Research Office (\$30,000). Principal Investigator for *Canadian Climate Policy Partnership: 2023-2026*.

University of Calgary, Mobilizing Alberta Climate Action Grant (\$59,880). Principal Investigator for *Empowering and including lower income households in the benefits of climate programming: 2023-2024*.

Clean Economy Fund (\$60,000). Principal Investigator for *Research on Inactive Wells in Canada: 2023*.

National Research Council (\$48,813). Principal Investigator for *Examining the Interplay between Building Codes and FPT Policies, Programs and Targets Related to GHG Emissions: 2022-2023*.

Mitacs Accelerate (\$45,000). Principal Investigator for *Developing a Database of Canadian Climate Policy and Programs: 2022-2023*.

Mitacs Business Strategy Internship (\$15,000). Principal Investigator for *A Techno-economic assessment of renewable natural gas (RNG) as a potential low carbon intensity fuel alternative eligible under Canada’s Clean Fuel Standard: 2022*.

Government of British Columbia (\$25,000). Principal Investigator for *Literature Review of Theories and Concepts related to Environmental Regulation: 2022*.

Smart Prosperity Institute (\$10,000). Principal Investigator for *Environmental Taxes and Energy Poverty in Canada: 2021*.

Calgary Foundation (\$35,000). Principal Investigator for *Multiple Account Benefit-Cost Analysis of Coal Mining in Alberta: 2021*.

University of Calgary Global Research Initiative in Sustainable Low Carbon Unconventional Resources (\$150,000), Principal Investigator for *Environmental Impacts of Oil and Gas Development: 2020-2023*.

Alberta Innovates (\$71,875). Principal Investigator for *An Analysis of Policy Mechanisms to Support Commercial Deployment of Bitumen Partial Upgrading in Alberta: 2020-2021*.

Emissions Reduction Alberta (\$15,500). Principal Investigator for *A Review of Barriers to Full-Scale Deployment of Emissions-Reduction Technologies*: 2019.

SSHRC Knowledge Synthesis Grant (\$30,000). Principal Investigator for *Public interest determination in environmental decision-making*: 2019-2020.

University of Calgary Global Research Initiative in Sustainable Low Carbon Unconventional Resources (\$181,000), Principal Investigator for *Methane Emissions Policy and Regulation in Canada: Assessing Options and Effectiveness*: 2019-2023.

Smart Prosperity Institute (\$10,000), Principal Investigator for *Canadian Resource and Environmental Economics Association (CREEA) PhD and ECR Workshop*: 2019.

University of Calgary Faculty of Arts Scholarly Activity Grant (\$1,000), Principal Investigator for *Canadian Resource and Environmental Economics Association (CREEA) PhD and ECR Workshop*: 2019.

Government of British Columbia (\$35,000), Principal Investigator for *Basic Income, Basic Services and Energy Transition*: 2018-2020 (project grant).

SSHRC Partnership Grant (\$2,500,000), Co-Investigator for *Greening Growth Partnership*: 2018-2022. PI: Stewart Elgie, University of Ottawa.

Smart Prosperity Institute (\$48,000), Principal Investigator for *Carbon Pricing Costs for Households and Revenue Recycling Options in Canada*: 2018-2020 (project grant).

Alberta Innovates (\$74,000), Principal Investigator for *Regulation of Partial Upgrading in Alberta*: 2018-2019

Government of Canada (\$10,000), Principal Investigator for *Best practices in Regulation Development & Management*: 2018

SSHRC Partnership Development Grant (\$200,000), Principal Investigator for *Canadian Northern Corridor*: 2018-2020.

University of Calgary Vice President of Research Office (\$10,000), Principal Investigator for *Canadian Northern Corridor*: 2018-2020.

School of Public Policy, University of Calgary: Strategic Research Initiative (\$30,000), Principal Investigator for *Canadian Northern Corridor*: 2018-2019.

School of Public Policy, University of Calgary: Strategic Research Initiative (\$5,000), Principal Investigator for *End of Life for Oil and Gas Wells*: 2018-2019.

University of Calgary Faculty Seed Grant (\$15,000), Principal Investigator for *Canadian Methane Policy and Regulation*: 2018-2019.

Protection of the Arctic Marine Environment (\$15,000), Principal Investigator for *Meaningful Engagement of Indigenous Peoples and Local Communities in Marine Activities*: 2018.

School of Public Policy, University of Calgary: Strategic Research Initiative (\$40,000), Principal Investigator for *Canadian Methane Policy and Regulation*: 2017-2018.

School of Public Policy, University of Calgary: Strategic Research Initiative (\$45,000), Principal Investigator for *Canadian Northern Corridor*: 2017-2018.

School of Public Policy, University of Calgary: Strategic Research Initiative (\$50,000), Principal Investigator for *Canadian Emissions Policy*: 2017-2018.

NSERC CREATE (\$1,650,000), Co-Investigator for *Responsible Development of Low-Permeability Hydrocarbon Resources (the ReDeveLoP Challenge)*: 2017-2024. PI: David Eaton, University of Calgary.

Alberta Innovates (\$50,000), Principal Investigator for *Partial Upgrading Impact Study*: 2016.

SSHRC Knowledge Synthesis Grant (\$25,000), Principal Investigator for *Indigenous, Industry and Government Perspectives on Consultation in Resource Development*: 2016-2017.

SSHRC Knowledge Synthesis Grant (\$25,000), Principal Investigator for *Hydraulic Fracturing and Public Policy*: 2015-2016.

SSHRC Insight Development Grant (\$40,810), Collaborator for *Internal Trade and Aggregate Productivity*: 2012-2015. PI: Trevor Tombe, University of Calgary.

FUNDS SECURED FOR THE SCHOOL OF PUBLIC POLICY

Government of Alberta (\$1,665,000), *Canadian Northern Corridor*: 2019-2023.

Alberta Real Estate Foundation (\$120,000), *Airbnb and the Second-hand Economy* and *Managing the End of Life of Oil and Gas Wells in Alberta*: 2019-2021.

Canadian Association of Petroleum Producers (\$50,000), *Energy for Life Program*: 2018.

Canadian Energy Pipeline Association (\$50,000), *Student Research Fellowships*: 2017-2021.

Alberta Innovates (\$40,000), *Alternative Emissions-Accounting Project*: 2016-2017.

Government of Alberta (\$30,000), Advisory services to the Energy Diversification Advisory Committee, 2017.

HONOURS AND AWARDS

[2023 Vanderkamp Prize](#) for the best article in *Canadian Public Policy*, joint with Brett Dolter and G. Kent Fellows for “Carbon Pricing Costs for Households and Revenue Recycling Options in Canada.” *Canadian Public Policy* 49(1): 13-45.

Canada’s Top 40 Under 40, 2021 (nominated)

Canada’s Top 40 Under 40, 2020 (nominated)

Canada Clean50 and Canada Clean16, 2019

Avenue Magazine, Calgary Top 40 Under 40, 2017

Alberta Oil Magazine, Top 35 under 35 in the Energy Sector, 2016

Young Women in Energy Award, 2014

PROFESSIONAL AND COMMUNITY SERVICE

Service

Treasurer, Canadian Resource and Environmental Economics Association, 2016 – present

Selection Committee, Bank of Canada Undergraduate Student Paper Awards, Canadian Economics Association Annual Meeting, 2022, 2023, 2024

Director, Canadian Society for Evolving Energy (formerly Canadian Society for Unconventional Resources), 2020 – 2023.

Section Editor (Energy and Environmental Policy), *The School of Public Policy Publications*, 2015 – 2022.

Member, City of Calgary Climate Panel, 2018 – 2024

Advisory Committee, Alberta Narratives Project, 2018

Policy Advising

Departmental Science Advisor, Environment and Climate Change Canada, Government of Canada, 2023
– present (60% interchange)

Visiting Scholar, Environment and Climate Change Canada, Government of Canada, 2022-2023
(sabbatical).

Prepared report (with William Scott, Andrew Pape-Salmon, Alaz Munzur & Rahim Mohamed),
*Examining the Interplay between Building Codes and Canadian FPT Policies, Programs, and
Targets Related to Reducing GHG Emissions*, for the National Research Council of Canada, 2023.

Advisor, Ministry of Environment and Climate Change Strategy, Government of British Columbia, 2022.

Advisor, Ministry of Energy, Mines and Low Carbon Innovation, Government of British Columbia, 2021-
2022.

Prepared report (with Sarah Dobson, Gregory Galay, Julia Jachimowicz, and W. Riley Sample),
Literature Review of Theories and Concepts related to Environmental Regulation, for the
Government of British Columbia, 2022.

Expert testimony, Oil sands emissions cap, House of Commons of Canada, Standing Committee on
Natural Resources, 2022.

Expert testimony, briefing note (with Megan Bailey, Sarah Dobson, G. Kent Fellows and Alaz Munzur),
*Output Based Pricing System Regulations Review: Comment from The Energy and Environmental
Policy Area at The School of Public Policy, University of Calgary*, submitted to Environment and
Climate Change Canada, 2022.

Prepared report (with Nancy Olewiler), *A Review and Assessment of the Natural Gas Royalty System in
British Columbia*, for the Government of British Columbia, 2021.

Expert testimony, briefing note (with Megan Bailey and Blake Shaffer), *Investment Tax Credit for
Carbon Capture, Utilization, and Storage Consultation Response*, submitted to Finance Canada,
2021.

Prepared report (with G. Kent Fellows and Alaz Munzur), *An Analysis of Industrial Policy Mechanisms to
Support Commercial Deployment of Bitumen Partial Upgrading in Alberta*, for Alberta Innovates,
2021.

Expert testimony, briefing note, “Potential Actions to Enable Carbon Pricing and Reduce Perverse
Subsidies,” prepared for HRH The Prince of Wales, 2020.

Prepared report (with Blake Shaffer), *Defining and Describing Energy Poverty in British Columbia: The
Distribution of Households’ Energy Expenditure*, for the BC Basic Income Panel, 2020.

Prepared report (with G. Kent Fellows and Victoria Goodday), *A Review of Barriers to Full-Scale
Deployment of Emissions-Reduction Technologies*, for Emissions Reduction Alberta, 2019.
(Subsequently published.)

Prepared report (with Akash Asif), *Comprehensive Assessment of Regulatory Programs*, for Treasury
Board of Canada Secretariat, Government of Canada, 2018.

Prepared report (with Brendan Boyd and Jessie Arthur), *Meaningful Engagement of Indigenous Peoples
and Local Communities in Marine Activities: Phase 2 Analysis*, for the PAME Working Group of the
Arctic Council, 2018.

Advisor, Federal Initiatives on Indigenous Relations, Government of Alberta, 2018.

Advisor to the Energy Diversification Advisory Committee, Government of Alberta, 2017.

Expert testimony, briefing note, “The Effect of Carbon Pricing on Canadian Households,” for the Senate of Canada, Standing Committee on Energy, the Environment and Natural Resources, 2017.

Expert testimony, appearance before the Senate of Canada, Standing Committee on Energy, the Environment and Natural Resources, 2017.

Member, Oil Sands and Crude Oil and Liquids working groups, Alberta Royalty Review, 2015-2016.

Prepared report (with Michal C. Moore), *The Role of Interties in the Alberta Market* for the Independent Power Producers Society of Alberta, 2011.

Refereeing

I have refereed 81 items in the period 2015-2024. The following is a list of journals and organizations I have assisted.

Alberta Law Review; *The B. E. Journal of Macroeconomics*; Bishop’s University (tenure file); Canada Research Coordinating Committee (New Frontiers in Research Fund); Canadian Climate Institute; *Canadian Journal of Agricultural Economics*; *Canadian Journal of Economics*; *Canadian Public Policy*; C. D. Howe Institute; *Climate Policy*; *Contemporary Economic Policy*; Dutch Research Council (NWO Talent Programme); *Energy Policy*; *Energy Research & Social Science*; *Environment, Development and Sustainability*; *The Extractive Industries and Society*; Frontier Centre for Public Policy; Government of Alberta; Hong Kong Research Grants Council; *International Journal of Environmental Research and Public Health*; IQ Carbone; *Journal of Environmental Economics and Management*; *Journal of Economic Dynamics and Control*; Max Bell Foundation; Mitacs (Accelerate Program); National Killam Program (Dorothy Killam Fellowship); Natural Resources Canada; Natural Sciences and Engineering Research Council (Arthur B. McDonald Fellowships); *Nature Communications*; *Politics & Governance*; Routledge Press; *Regulation & Governance*; *Review of Policy Research*; Social Sciences and Humanities Research Council (Insight Grants); *The School of Public Policy Publications* (University of Calgary); Taylor and Francis Press; University of Victoria (tenure files).

Conference Selection Committees

CREEA 2024 Early Career Researcher Workshop; CREEA 2023 Early Career Researcher Workshop; CREEA 2019; CREEA 2019 Early Career Researcher Workshop; CREEA 2018; CEA 2018; CEA 2017; CEA 2016; CREEA 2016.

DEPARTMENT/SCHOOL AND UNIVERSITY SERVICE

2024 –	Member, Strategic University Proposal and Platform Opportunity Review Team (SUPPORT): Research Awards Committee, University of Calgary
2023 –	Member, Equity, Diversity, Inclusion and Accessibility and Indigenous Engagement Committee, School of Public Policy, University of Calgary
2023 – 2026	Member (Arts’ Dean’s Representative), General Faculties Council, University of Calgary
2015 –	Member, Research Planning Committee, School of Public Policy, University of Calgary
2022 – 2023	Member, Search Committee, Director, School of Public Policy

2022 – 2022	Member, Search Committee, MPP Program Director, School of Public Policy
2021 – 2022	Member, General Faculties Council Executive, University of Calgary
2021 – 2022	Member, Research Committee, Department of Economics, University of Calgary
2021 – 2022	Member, Search Committee, Chemical and Petroleum Engineer, Schulich School of Engineering
2021 – 2021	Member, Selection Committee, Academic Director, School of Public Policy
2020 – 2022	Member, General Faculties Council, University of Calgary
2020 – 2022	Member, Energy Research Strategy 2 Working Group
2020 – 2022	Member, Strategic University Proposal and Platform Opportunity Review Team (SUPPORT): Partnerships and Major Grants Committee, University of Calgary
2020 – 2021	Member, Search Committee, Operations and Supply Chain Management, Haskayne School of Business
2016 – 2022	Member, Engagement Committee, School of Public Policy, University of Calgary
2015 – 2022	Member, Executive Committee, School of Public Policy, University of Calgary
2017 – 2021	Member, Policy & Planning Committee, Department of Economics, University of Calgary
2015 – 2021	Member, Admissions Committee, School of Public Policy, University of Calgary
2017 – 2019	Graduate Program Committee, School of Public Policy, University of Calgary

INVITED CONFERENCE PRESENTATIONS, SEMINARS AND PANELS

- 2024 Philippines-Canada Forum on Energy Transition, Global Affairs Canada, Manila (planned); Canadian Resource and Environmental Economics Association 34th Annual Meeting, Kelowna; Canada's Productivity Summit, School of Public Policy, University of Calgary, Calgary, AB (moderator); Workshop on SMEs in Asia, School of Public Policy, University of Calgary, Calgary, AB; 2024 Air & Waste Management Association (AWMA) Annual Conference, Calgary (withdrew); GeoConvention 2024 Energy Innovation Pavilion; Canadian Economics Association 58th Annual Meeting, Toronto (keynote; Environment and Climate Change Canada Special Lecture); Academia-Government Research Exchange, Rotman School of Management, University of Toronto, Toronto, ON; Ontario Network for Sustainable Energy Policy 2024 workshop, Kingston, ON; Society of Petroleum Engineers, SPE Forum: Methane Emissions Management for a Sustainable Future, Banff, AB.
- 2023 The Professional Institute of the Public Service of Canada, 104th Annual General Meeting, Montreal; University of Manitoba, Department of Economics seminar series, Winnipeg; Canadian Resource and Environmental Economics Association 33rd Annual Meeting, Ottawa; CAMPUT Annual General Meeting, Whitehorse; Canadian Economics Association 57th Annual Meeting, Winnipeg (keynote; 50th Anniversary *Canadian Public Policy* Lectures); Atlanta Conference on Science and Innovation Policy, Atlanta, GA; REDEVELOP Annual Conference, Calgary; Canadian Biogas Association, 2023 Value of Biogas East Conference, Toronto (keynote); Forum on Carbon Neutrality in Atlantic and Eastern Canada (online); Climate Risk/Change Day Virtual Workshop, Risk and Insurance Studies Centre, York University (online); Carleton University, Department of Economics seminar series, Ottawa; Environment and Climate Change Canada, Visiting Scholar Speaker Series (online).

- 2022 Women and Inclusivity in Sustainable Energy Research, Policy Change, Energy Democracy & Inclusive Energy Policies (online); University of Calgary, Global Development Society and International Relations Students Association, Career Advice: Working at the Government of Canada, Calgary; Chevron DA DQ Webinar Series (online); Canadian Resource and Environmental Economics Association Early Career Researcher Workshop (keynote), Ottawa; Canadian Economics Association Annual Meeting, Ottawa; School of Public Policy and Global Affairs Canada, Trade Experts Roundtable, Ottawa; Instituto Español de Estudios Estratégicos and Repsol, Energy and Geostrategy 2022, Madrid; University of Alberta, Department of Resource Economics and Environmental Sociology seminar series (student choice); Gas Alberta Inc. Board Meeting; Society of Decision Professionals, Calgary Chapter (webinar); Brookings Institute, Can state-level carbon pricing help achieve national climate goals? (webinar); Coalition of Finance Ministers for Climate Action, HP3 Workshop on Pan-Canadian Approach to Pricing Carbon Pollution (webinar).
- 2021 World Petroleum Council, 23rd World Petroleum Congress, Houston; Royal Society of Canada, G7 Research Summit | Climate Change Implications, Adaptation and Mitigation Options for Canada (online); University of Calgary, Community Engagement Webinar: The Road to Net-Zero (webinar); Department of Economics, University of Victoria; Alberta Institute of Agrologists; Columbia University, Columbia Center on Global Energy Policy, Exploring an Uncertain Energy Future: Equinor Energy Perspectives 2021 (webinar); University of Calgary, Idea Exchange: Engineering Thriving Cities (webinar); School of Public Policy, University of Calgary, A House Divided: Results of the 2021 Federal Election (webinar); GeoConvention 2021 (online); School of Public Policy, University of Calgary, Alberta's Future: Pathways Forward for Energy and the Environment (webinar); Max Bell School of Public Policy, McGill University and Environment and Climate Change Canada, Workshop: Interactions between OBPS and other compliance markets (online); Society of Petroleum Engineers, SPE Workshop: CCUS in Canada – Opportunities for a Net-Zero Future (webinar); World Petroleum Council, Virtual Expert Workshop: Sustainable Strategies for HSE in the Oil, Gas & Petrochemical Industries (webinar); Johnson Shoyama Graduate School of Public Policy, Budget Day 2021: COVID-19, Deficits and the Impacts on Western Canada (webinar); Canadian Heavy Oil Association, Leading Toward Net Zero in Heavy Oil (webinar); Women of #ABpoli Podcast; Alberta Electric System Operator, Economic Efficiency Forum; University of Alberta China Institute, Impacts of Biden's Climate Change Policy on China and Canada (webinar); Parliamentary Internship Program, Climate and Energy Policy in Canada and the United States (webinar).
- 2020 CBC, West of Centre Podcast; HRH The Prince of Wales' Sustainable Markets Initiative, Roundtable on Carbon Pricing and Perverse Subsidies Schemes (online); Online Agricultural and Resource Economics Seminar (webinar); Canadian Resource and Environmental Economics Association 30th Annual Conference (webinar); International Association of Energy Economics Webinar Series (webinar); GeoConvention 2020 (webinar); YPAC x Avatar Program (webinar); C. D. Howe Institute, COVID-19 Working Group on Business Continuity and Trade (online); Emissions Reduction Alberta, Alberta's Energy Future (webinar); European Association of Environmental and Resource Economists 25th Annual Conference (webinar).

- 2019 CBS Inequality Platform's 1st Workshop on Climate Change and Inequality, Copenhagen (withdrew); Atlantic Canada Economics Association, Annual Conference, Moncton; energy3 – Canada's Energy Conference, Halifax; Environment and Climate Change Canada, Carbon Pricing Workshop; Canadian Resource and Environmental Economics Association 29th Annual Conference, Charlottetown; World Petroleum Council 6th Future Leaders Forum, St. Petersburg, Russia; Congress of the Humanities and Social Sciences and Canadian Political Science Association, Annual Conference, Vancouver; Canadian Economics Association, Annual Conference, Banff; Environment and Climate Change Canada, Canada-France Seminar on Carbon Pricing, Paris; Instituto Español de Estudios Estratégicos and Repsol, Energy and Geostrategy 2019, Madrid; McGill Institute for the Study of Canada Annual Conference, Montreal; Global Affairs Canada Minneapolis Consulate, Energy Network Meeting; The Canadian Institute, 19th Annual Arctic Oil & Gas Symposium, Calgary; Calgary Association of Lifelong Learners, Science and Environment Speaker Series, Calgary; Canadian Society of Exploration Geophysicists, Technical Luncheon, Calgary.
- 2018 School of Public Policy, Measuring the Impacts of Energy Infrastructure, Ottawa; Haskayne School of Business, Haskayne Hour, Opportunities Down the Pipe: Influences on successful Indigenous and industry relationships, Calgary; Council for Clean and Reliable Energy, Energy Leaders Roundtable, Kananaskis; University of Calgary, Sustainable Development Goals Summit: Local Action, Global Impact (keynote), Calgary; Yukon First Nations Chamber of Commerce, Arctic Indigenous Investment Conference, Yellowknife; Global Petroleum Show 2018, Calgary; Ecofiscal Commission, School of Public Policy and the Max Bell School of Public Policy, Bridging Divides: In Search of Sound Public Policies for Energy and Environment in Canada, Calgary; Canadian Society for Unconventional Resources, Fractured Ground: Hydraulic Fracturing and Public Policy, Calgary; Treasury Board Secretariat of Canada; Regulatory Review Knowledge Synthesis Roundtable, Ottawa; University of Calgary, Calgary and Beyond: Sustainability Next 20 Years speaker series, Calgary; University of Calgary, Department of Geoscience, Friday Afternoon Talk Series, Calgary; Thompson Rivers University, School of Business and Economics Seminar Series, Kamloops.
- 2017 Mowat Centre, Confederation of Tomorrow 2.0, Toronto; Delphi Group, Energy Sector Sustainability Leadership Initiative workshop, Calgary; University of Toronto School of Public Policy and Governance, Canada's Policy Transformations: The Last 50 Years and the Next 50, Toronto; Energy In Society, Global Energy Transition Workshop, Calgary; SSHRC, Forum on Indigenous, Industry and Government Perspectives on Consultation in Resource Development, Ottawa; Global Petroleum Show, Low Carbon Innovation Forum, Calgary; Atlantic Canada Economics Association Annual Meeting, Antigonish; APEGA Calgary Branch Luncheon, Calgary; CAMPUT: Canada's Energy and Utility Regulators, CAMPUT Annual Conference, Vancouver; Canadian Heavy Oil Association, Technical Panel Presentation, Calgary; Webber Academy, The President's Breakfast Club ATB Financial Speaker Series, Calgary; Alberta Beef Industry Conference, Red Deer; Senate of Canada, testimony to Standing Committee on Energy, the Environment and Natural Resources (webcast), Calgary; University of Calgary, Faculty of Arts Speakeasy, Calgary; University of Calgary, Department of Geoscience, Friday Afternoon Talk Series, Calgary.

- 2016 Canadian Society of Petroleum Geologists Environment Division, lunch and learn series, Calgary; The Energy Roundtable, 2016 Calgary Energy Roundtable, Calgary; International Right of Way Association Chapter 48, IRWA Chapter 48 Education Symposium, Calgary; The Canadian Institute, Environmental Law and Regulation conference, Calgary; Canadian Independent Petroleum Marketers Association, Western Members Meeting (keynote), Calgary; Arctic Energy and Emerging Technologies Conference and Tradeshow, Inuvik; Energy and Mines Ministers' Conference, National Workshop on Public Confidence in Energy and Mining Development, Winnipeg; Alberta Utilities Commission, Energy Regulatory Forum, Calgary; CAMPUT: Canada's Energy and Utility Regulators, CAMPUT Annual Conference, Montreal; Canadian Prairie & Northern Section Air & Waste Management Association, CPANS Annual Conference, Calgary; Runnymede Society, Keystone XL: Pipelines, Policy & Politics, Calgary; Ecofiscal Commission, Federal Provincial Carbon Coordination webinar; Economics Society of Calgary, Alberta's Climate Leadership Plan and Canada's Paris Commitments: Implications for the Provincial Economy its Competitiveness.
- 2015 SSHRC and Natural Resources Canada, Knowledge Symposium on Energy & Natural Resources, Ottawa; Calgary Chamber of Commerce, How will global markets and FX rates affect your business in 2016?, Calgary; Energy Council of Canada, Energy Summit, Toronto; Biorefining Conversions Network, Green Chemistry Workshop, Edmonton; Canada Science and Technology Museums Corporation, Let's Talk Energy webinar; Chartered Professional Accountants, Canada Tax Symposium West (keynote), Calgary; National Public Relations LNG Forum, Ottawa; Canadian Political Science Students' Association National Conference, Calgary.
- 2014 Israel Defense: Energy 2015, Tel Aviv; 2nd Annual Canadian NGL Markets, Infrastructure & LPG Exports, Calgary; Petroleum Acquisition and Divestitures Association, Calgary; Alberta GHG Summit, Calgary; Canadian Resource and Environmental Economics Study Group Annual Conference, Saskatoon; Brookings Institution Natural Gas Task Force, Washington D.C.; BP Canada Energy Group Customer Forum, Banff; Alberta's Energy Future (panel organized by the Liberal Party of Canada), Calgary; University of Calgary, Department of Economics Alumni Conference; Canadian Economics Association Annual Meeting, Vancouver.
- 2013 Energy Literacy Directions, Kananaskis Village; Canadian Economics Association Annual Meeting, Montreal; World Petroleum Council Youth Forum, Calgary.
- 2011 Sonoma State University, Department of Economics Seminar, Sonoma; Missouri University of Science and Technology, Rolla; Department of Economics Seminar; Oklahoma State University, Spears School of Business Seminar, Stillwater; University of Alberta, School of Business Seminar, Edmonton; Canadian Economics Association Annual Meeting, Ottawa.
- 2010 University of Calgary, Department of Economics Seminar; Canadian Resource and Environmental Economics Study Group Annual Meeting, Guelph; University of Calgary, Department of Economics Workshop.

- 2009 University of Calgary, Department of Economics Summer Seminar Series; Petroleum History Society of Calgary June Meeting; University of Calgary, Department of Economics Graduate Students Mini-Conference; Canadian Economics Association Annual Meeting, Toronto.
- 2008 Canadian Economics Association Annual Meeting, Vancouver.
- 2007 Queen's University Economic History Workshop, Kingston; University of Calgary Department of Economics Workshop.

CONFERENCE ORGANIZATION

Canadian Resource and Environmental Economics Association 34th Annual Conference, Kelowna, 2024
 Canadian Resource and Environmental Economics Association 33rd Annual Conference, Ottawa, 2023
 Canadian Resource and Environmental Economics Association 9th Canadian PhD and Early Career Researcher Workshop, Winnipeg, 2023
 The Environmental and Community Impacts of Inactive Oil and Gas Wells Workshop, School of Public Policy, University of Calgary (online), 2023
 Canadian Resource and Environmental Economics Association 30th Annual Conference (online), 2020
 Canadian Resource and Environmental Economics Association 6th Canadian PhD and Early Career Researcher Workshop, Calgary (online), 2020
 Canadian Resource and Environmental Economics Association 5th Canadian PhD and Early Career Researcher Workshop, Calgary, 2019
 The School of Public Policy, Measuring the Impacts of Energy Infrastructure, Ottawa, 2018
 The School of Public Policy, Indigenous Policy Conference: Beyond Reconciliation, Calgary, 2016
 Canadian Resource and Environmental Economics Study Group Annual Conference, Banff, 2016
 The School of Public Policy, Canadian Network for Energy Policy Research and Analysis Conference on Public Acceptance of Energy Projects, Ottawa, 2015
 The School of Public Policy, Canadian Network for Energy Policy Research and Analysis Workshop, Calgary, 2015
 The School of Public Policy, Labour Markets Roundtable, Calgary, 2014
 The School of Public Policy, Social Licence Symposium, Calgary, 2014
 The School of Public Policy, Regulatory Roundtable: Social Licence, Denver, 2014
 The School of Public Policy, Risk and Its Influence on Public Policy Symposium, Calgary, 2014
 The School of Public Policy, Asian Market Diversification Symposium, November 2013
 The School of Public Policy, Energy Regulation Symposium, Calgary, 2013
 The School of Public Policy, Regulatory Roundtable: Regulatory Hurdles, Denver, 2013
 The School of Public Policy, Dutch Disease Symposium, Toronto, 2013

CONFERENCE ATTENDANCE

We Did It! A Future History of Net Zero in Canada, Environmental Governance Lab, University of Toronto, 2022; Canadian Resource and Environmental Economics Association Annual Conference, 2022; Canadian Resource and Environmental Economics Association Annual Conference, 2021; Canadian Resource and Environmental Economics Association Annual Conference, 2018; Canadian Economics Association Annual Meeting, 2018; Canadian Resource and Environmental Economics Study Group Annual Conference, 2017; Canadian Economics Association Annual Meeting, 2017; American Economics Association Annual Meeting, 2016; Canadian Resource and Environmental Economics Study

Group Annual Conference, 2016; Canadian Resource and Environmental Economics Study Group Annual Conference, 2015; Canadian Economics Association Annual Meeting, 2015; Positive Energy Conference, 2015; American Economics Association Annual Meeting, 2015; Canadian Resource and Environmental Economics Study Group Annual Conference, 2014; RBN School of Energy, 2014; Canadian Resource and Environmental Economics Study Group Annual Conference, 2013; Rocky Mountain Empirical Trade Conference, 2013; Canadian Energy Research Institute Natural Gas Conference, 2013; American Economics Association Annual Meeting, 2013; Canadian Resource and Environmental Economics Study Group Annual Conference, 2012; Canadian Economics Association Annual Meeting, 2012; Canadian Resource and Environmental Economics Study Group Annual Conference, 2009; University of Calgary School of Public Policy Inaugural Conference, 2009; University of Calgary Institute for United States Policy Research Transboundary Water Policy Issues: The Western North American Region, 2008; University of British Columbia Summer Conference on Industrial Organization, 2006.

TEACHING EXPERIENCE

University of Calgary, MPP Capstone Course (PPOL 620/623) – Spring/Summer 2025; Spring/Summer 2024.

University of Calgary, Responsible Development of Hydrocarbons and Policy (PPOL 611.23/699.02) – Spring 2022 (hybrid); Spring 2021 (online); Spring 2020, Spring 2019

University of Calgary, Foundations I: Analytical Foundations (PPOL 601) – Fall 2020 (co-taught with Dr. Lindsay Tedds)

University of Calgary, The Multi-Faces of Israel (PPOL 611.07) - Winter 2020 (instructor of record); Winter 2019, Winter 2018 (on-location faculty supervisor for study abroad course)

University of Calgary, Energy and Environmental Policy (PPOL 611.02) - Winter 2015, Fall 2016, Fall 2017, Fall 2018

University of Calgary, Decision Analysis (PPOL 609) - Winter 2016, Winter 2017, Winter 2018

University of Calgary, Intermediate Macroeconomics (ECON 303) - Spring 2010

University of Calgary, Introductory Microeconomics (ECON 201) - Summer 2009

Teaching Assistant, University of Calgary, 2005 – 2011

Guest Lectures

UBC, Allard School of Law, Tax Policy – Winter 2023.

University of Calgary, The Study of Public Policy (PPOL 600) – Fall 2020 (online); Fall 2021

University of Calgary and Laval University, Summer School on the Geopolitics of Natural Resources (PPOL 611.21) - Spring 2014, Spring 2015, Spring 2016, Spring 2019

University of Calgary, Dentons' International Energy Law Study Tour (Law 693) - Winter 2017, Winter 2018, Winter 2019, Fall 2019

ACADEMIC SUPERVISION

Postdoctoral Scholars

Tetiana Semeniutina, 2022 – present

Katharina Koch, 2020 – 2023; placement: School of Public Policy, University of Calgary

Gregory Galay, 2021 – 2023; placement: Environment and Climate Change Canada

Evgeniia (Jen) Sidorova, 2021 – 2022; placement: Pauktuutit Inuit Women of Canada

Brendan Boyd, 2016 – 2017; placement: MacEwan University

E. Dylan Morgan, 2015; placement: Environment and Climate Change Canada

PhD Students

Coleman Vollrath, University of Calgary, 2024 (PhD Candidacy Examiner)

Adegboyega During, PhD – University of Calgary, *Essays in Energy and Policy Analysis*, 2019 (PhD Defence Internal Examiner)

Zainab Dadashi Forshomi, University of Calgary, *Evaluating Oil Sands Technologies Using Life Cycle Assessment and Mathematical Optimization*, 2017 (PhD Candidacy External Examiner)

Julie Williams, PhD – University of Victoria, *Environmental Leadership and Regulation: Policy Implications for Provincial Governments in Canada*, 2015 (PhD Defence External Examiner)

Master's Students

Jennifer Rossi, MPP Candidate, 2023-2024

Dylan Rama, MPP, *The Emissions Reduction Effects of Emission Pricing Systems: A Structured Literature Review*, 2023-2024

Bodgan Revnic, MPP, *The Digital Divide: Federal Broadband Strategy and Implications for Indigenous Communities in Rural Canada*, 2023-2024 (co-supervisor)

Taylor Robinson, MPP, *How does Alberta's K-12 Education Funding Model Impact Student Funding Outcomes?*, 2023-2024

Lindsay Geier, MPP, *Distributional Effects of Changes in BC's Carbon Tax Revenue-Use*, 2021-2022

Julia Jachimowicz, MPP, *Reducing Emissions, Not Cows: Regulatory Policies to Mitigate Methane*, 2021-2022

Jordan Williams, MSc SEDV, *An economic assessment of renewable natural gas (RNG) as a potential low carbon intensity fuel alternative eligible under Canada's Clean Fuel Regulations*, 2021-2022

Kimberly Markvoort, MSc SEDV, *Assessment of Indigenous Perspectives embodied in Sustainability Reporting and ESG Disclosure Practices of Canadian Exploration and Production Companies*, 2020-2022

Coleman Vollrath, MSc, *Methane Emissions from the Global Oil and Gas Industry: A Scoping Review to Characterize the Research Trends, Knowledge Gaps, and Priorities*, 2020-2022 (committee member)

Mehdi Salamati, MSc, *Optimal Routing of Multi-Modal Wide Energy and Infrastructure Corridors*, 2022 (committee member)

Amelia Harman, MPP/JD, *Assessing the Performance of Aboriginal Financial Institutions Against Mainstream Banks and Credit Unions*, 2020-2021

Keshia Holloman-Dawson, MPP/JD, *The Methane Regulation Gap: Regulation of Abandoned Oil and Gas Wells to Assist Methane Emission Reduction Targets in Alberta*, 2020-2022

Andreas Murphy, MPP, *A Case Study in Government Mismanagement of Delayed Hydroelectric Megaprojects: The Site C Perspective*, 2020-2021

Emma Dizon, MPP, *The Impact of the Regulatory Environment on the Ease of Doing Business: How Calgary Measures Up*, 2019-2020

Laura Flinkfelt, MPP, *Evaluating Toronto's Waste Diversion and Generation Performance 2006–2019*, 2019-2021

Christian Loudon, MPP, *Measuring and Mitigating CleanBC's Impact on Energy Affordability*, 2018-2020

Marcus Threndyle, MPP/JD, *Developing renewable electricity generation with feed-in tariff policy*, 2019-2020

Alyssa Anderson, MPP, *Alberta in the Age of Renewable Power: Policy Lessons from Germany and Sweden*, 2018-2019

Mitchell Boyne, MPP, *Oil Change: How the Global Oil and Gas Downturn Transformed Canada's Energy Industry*, 2018-2019

M. Shae Doyle, MPP, *Microgeneration and Smart Grids: The Alberta Context*, 2017-2019

Ryan Green, MPP, *Bending the Curve: Reducing Alberta's Oil & Gas Liabilities via Timelines on Well Abandonments*, 2018-2019

April Kargard, MPP, *Investigation into the Effects of Resource Development Projects on Rural and Indigenous Communities*, 2018-2019

Braeden Edwards, MPP, *Pipelines, Public Interest, and Regulatory Change: The Public Interest Implications of Including Indirect Greenhouse Gas Emissions in Federal Environmental Assessments*, 2017-2018

Neuczki Mathurin, MPP, *Drilling into Policy Effectiveness: An Evaluation of Carbon Pricing Policies in Oil and Gas Jurisdictions*, 2017-2018

Jade McLean, MPP, *Public Perceptions of Pipeline Development in Canada*, 2017-2018

Graeme Davidson, MPP candidate, 2017

Jessie Arthur, MPP, *Carbon Capture in Alberta: Costs, Benefits, and Policy*, 2016-2017

Kiran Gurm, MPP, *The feasibility of reducing Nunavut's diesel reliance with a transition to renewable energy technologies, primarily solar and wind energy*, 2016-2017

Karen Spencer, MPP, *Four Easy Ways to Reduce Canada's GHG Emissions*, 2016-2017

Austin Thompson, MPP, *Evaluating For-Profit Ridesharing Regulations in Canada*, 2016-2017

Zach Biggs, MPP, *How to Protect Lower-Income Albertans from a Carbon Tax*, 2015-2016

Brendan Frank, MPP, *Keeping the Lights On: Renewable Power in Alberta's Post-Coal Era*, 2015-2016

Beth McDonald, MPP, *Examining the Role of Electoral Systems in the Policy Influence of Aboriginal Populations: A Comparative Analysis of the Canadian, Australian, and New Zealand Cases*, 2015-2016

Laura Johnson, MPP, *Regulation of small modular nuclear reactors in Canada*, 2015

EMPLOYEE SUPERVISION

(Excludes part-time research assistants)

Ruba Khan, Research Coordinator, School of Public Policy, 2024 – present

Katharina Koch, Research Associate, School of Public Policy, 2023 – present

Kayla Arcand, Summer Research Assistant, School of Public Policy, 2021

Kalab Begna, Summer Research Assistant, School of Public Policy, 2018

Sarah Dobson, Research Associate, School of Public Policy, 2015 – 2022

Owen Duffy, Summer Research Assistant, School of Public Policy, 2024

G. Kent Fellows, Research Associate, School of Public Policy, 2015 – 2021

Emily Galley, Research Coordinator, School of Public Policy, 2020 – 2022

Louise Gilleard, Summer Research Assistant, School of Public Policy, 2022

Victoria Goodday, Research Associate, School of Public Policy, 2018 – 2021

Julia Jachimowicz, Summer Research Assistant, School of Public Policy, 2022

Andrea Jaros, Program Coordinator, School of Public Policy, 2021

Shantel Jordison, Manager, School of Public Policy, 2015 – 2016

Hannah Lederman, Program Coordinator, School of Public Policy, 2019 – 2020
Sophie Lorefice, Research Associate, School of Public Policy, 2014 – 2017
Kimberly Markvoort, Program Coordinator, School of Public Policy, 2022
Kathleen Mullrooney, Summer Research Assistant, School of Public Policy, 2019
Alaz Munzur, Research Associate, School of Public Policy, 2020 – 2022
Evelyn Taylor-McGregor, Summer Research Assistant, School of Public Policy, 2012

PROFESSIONAL AFFILIATIONS

American Economics Association (2011, 2013; 2015-2016); Canadian Economics Association (2010 – present); Canadian Resource and Environmental Economics Association (2010 – present); Smart Prosperity Institute Economics and Environmental Policy Research Network (2018 – present); Young Women in Energy (2014 – present); Women and Inclusivity in Sustainable Energy Research (2022 – present).

OTHER

Regularly provide solicited information, advice, and guidance to governments and public servants in Canada.

Regularly appear on television, radio, podcasts, print media, and other forums, providing economic and policy commentary on current events. This is typically 20-30 appearances in a given year, and higher in election years.

Various general-interest articles and op-eds in *Alberta Views*, *Calgary Herald*, *CBC Opinion*, *CBC News*, *Edmonton Journal*, *The Globe and Mail*, *The Hill Times*, JWN News, *Vancouver Sun*.

Blog posts on via the Canadian Climate Institute and the School of Public Policy, University of Calgary.